

ZapSortBot: Autonomous robotic sorting of used batteries utilizing deep learning

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Abstract— The increasing volume of spent batteries from consumer electronics and electric vehicles poses significant environmental and safety challenges for recycling systems. Efficient battery sorting by chemistry and size is critical for ensuring safe downstream processing and maximizing material recovery. This paper presents a low-cost, modular system that classifies and sorts batteries using a multi-modal approach combining AI-powered vision and weight-based inference. *ZapSortBot* employs a YOLO-v8 object detection model for real-time localization, with downstream logic determining battery chemistry type based on visual and physical cues. A 6-DoF robot (Niryo Ned 2) and a conveyor belt have been used to perform the sorting. The system achieved a classification accuracy of 98.3% across four common chemistries (alkaline, nickel-metal hydride, lithium, and zinc-carbon), with robust performance under varying lighting and battery conditions. Compared to high-cost industrial systems relying on spectroscopic techniques or X-ray imaging, *ZapSortBot* offers an accessible alternative for decentralized recycling environments. The system’s modularity, retainability, and special-case handling further support deployment in resource-constrained settings, addressing a critical gap in scalable, intelligent battery sorting technology.

I. INTRODUCTION

The widespread use of battery-powered devices (from consumer electronics to electric vehicles) has led to a surge in end-of-life batteries requiring safe, efficient recycling. These batteries, especially lithium-ion, nickel-metal hydride, and zinc-based types, contain both valuable metals and hazardous substances. Accurate sorting is essential to maximize recovery and reduce safety risks, yet manual methods remain inefficient due to chemical variability, poor labeling, and thermal hazards [1, 2, 8].

Although advanced systems using spectroscopy, X-ray imaging, and sensor fusion offer high accuracy [13, 14], their cost and infrastructure needs make them unsuitable for decentralized or small-scale operations. This highlights the need for low-cost, intelligent solutions that perform reliably in resource-constrained environments.

ZapSortBot addresses this gap through a multi-modal system combining real-time object detection and weight-based inference. Table I displays the comparisons between *ZapSortBot* and existing systems. As can be seen on table I, while systems relying on humans for sorting come with low to moderate setup costs compared to automated systems, they are error prone and also come with added labor costs [16]. Current automated sorting systems achieve high accuracy using technologies (LINEV, TOMRA...) such as

spectroscopy to be able to identify unlabeled or damaged batteries [13,14]. However, those advanced systems come with higher costs than manual or semi-automated sorting and unlike them *ZapSortBot* offers a scalable, low-cost approach to battery sorting in both research and applied engineering contexts.

This paper describes the *ZapSortBot* system and methodology (Section II), presents results (Section III), and finally summarizes the work and outlines future directions (Section IV).

TABLE I. COMPARISON OF EXISTING SYSTEMS

System Type	Technology Used	Chemistry Resolution	Special Case Handling	Cost & Accessibility
Manual Sorting	Human inspection, labels	Low (Manual visualization based)	Error-prone, unsafe	Low setup cost, high labor cost
Semi-Automated Sorting	Mechanical aids + human oversight	Moderate (flagging with human review)	Semi-flagging, still error-prone	Moderate cost and accessibility
Existing Fully Automated Systems (e.g. TOMRA, LINEV) [13,14]	CV, LIBS/XRT, robotics, AI	High (>98% with spectroscopy)	Adaptive with AI	Very high (industrial scale only)
ZapSortBot	Multi-modal sensing +Deep learning +Robotics	High (>98% for the test cases with normal camera)	Reasonably Robust with fallback logic	Low-cost, accessible for both small and large-scale use

II. METHODOLOGY

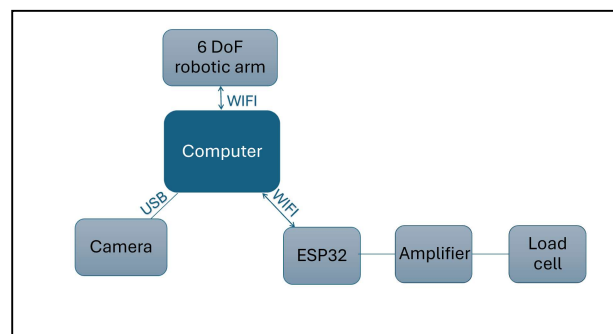


Fig. 1. ZapSortBot’s block diagram

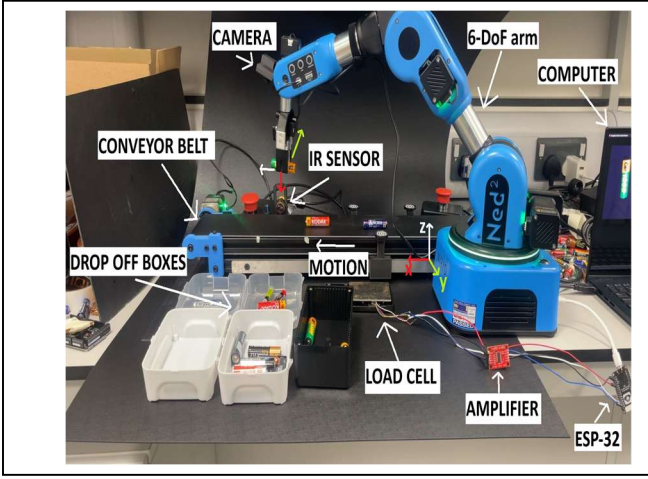


Fig. 2. System's Final Layout

The *ZapSortBot* system uses an algorithm designed to address the battery sorting as shown in the flowchart below:

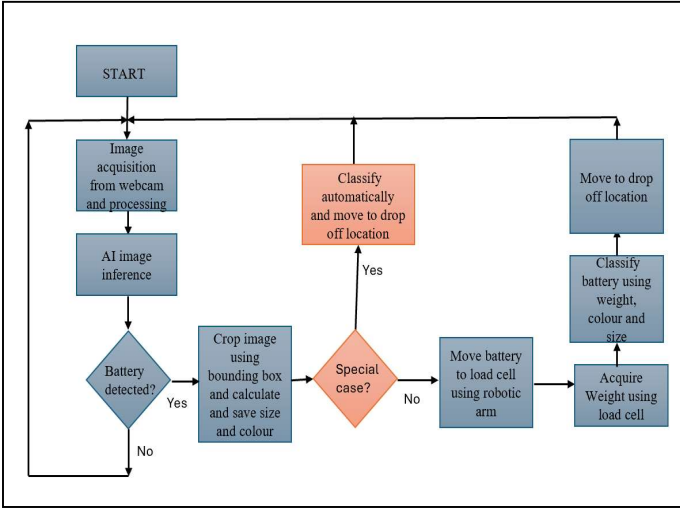


Fig. 3. ZapSortBot's System Flowchart

Hardware Setup

As displayed in figure 1, the system consists of a robotic arm, a camera, a computer controlling the entire process and a load cell with the necessary hardware (amplifier+ESP32). It also includes a motorized conveyor with an IR object that transports batteries from one end of the conveyor belt (left) to the other one (under the camera), halting automatically when an object is detected. Its matte-black surface ensures optimal contrast for visual detection. The robotic arm, equipped with a friction-enhanced gripper, transfers batteries for classification and sorting. Figure 2 displays the final setup with the labelled components.

Software Interface and Configuration

Initial communication was established via Ethernet through Niryo Studio, with manual wireless reconfiguration following a firmware update. A Python-based GUI (figure 4) was designed to enable real-time classification, dataset

management, live inference testing, and robot pose updates. No knowledge of programming is needed for the end user.

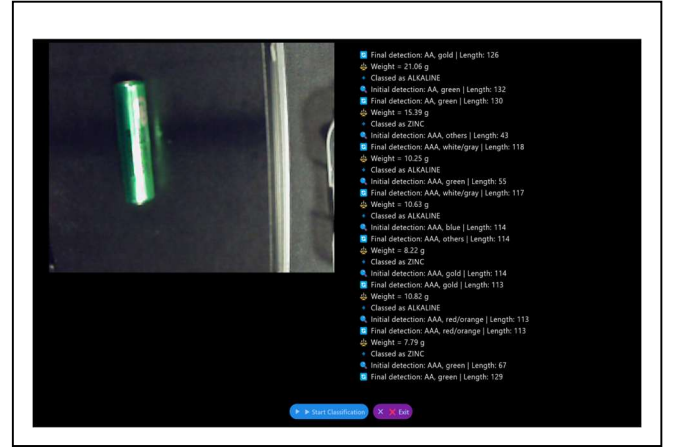


Fig. 4. ZapSortBot's GUI (Classification in progress)

Model Training

YOLOv8 model pre-trained on COCO, was fine-tuned using transfer learning with 80 custom battery (AA images (75/25 training/test)). Training used a frozen backbone, anchor-free design, and augmentations (flips, rotations, noise) to improve generalization.

Sensor Fusion and Classification Logic

As depicted in the flowchart (Figure 3), the algorithm initiates by capturing an image from the conveyor belt using a camera. This image is processed by a trained YOLOv8 model to detect the presence of a battery. If a match with a presaved special case battery (batteries that don't fit the normal characteristics of their chemistry) is found, the standard classification pipeline is bypassed, and a predefined label is assigned. If no match is identified, the battery is transferred to a load cell to measure its weight. Finally, the algorithm utilizes the collected features to classify the battery according to predefined logic.

Robotic Control

The Niryo Ned2 executes sorting via predefined poses: VIEW, PICK, WEIGHT_DROP, and *_DROP. Cartesian motion planning with inverse kinematics is used to move and position the robot for various maneuvering. The gripper can rotate to align with the direction of the battery. A one-second pause ensures stable weight readings. All steps are wrapped in error-handling logic, with status updates shown via the GUI.

III. RESULTS AND DISCUSSION

1) Training Results

The YOLOv8 object detection model was trained over 43 epochs using a custom-labelled dataset comprising images of AA and AAA batteries with diverse sizes, orientations, and colors. As displayed in figure 5, training convergence was evident through steadily declining loss values across the box, class, and Distribution Focal Loss (DFL) metrics for both

training and validation datasets, indicating effective learning of spatial and categorical features.

Final training losses:

- Box loss: 0.534
- Classification loss: 0.916
- Distance Focal Loss (DFL): 0.901

Validation performance metrics:

- Precision: 99.76%
- Recall: 100%
- mAP@0.5: 99.5%
- mAP@0.5:0.95: 74.1%

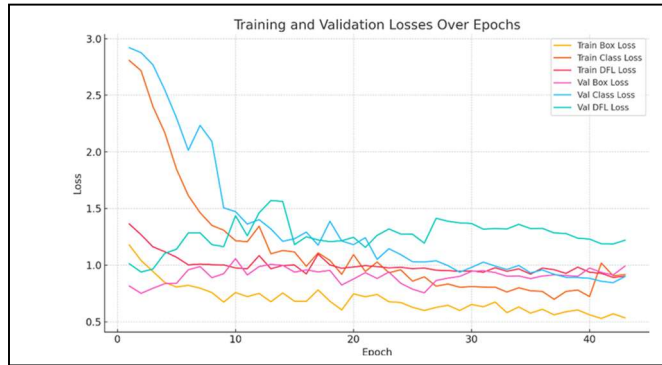


Fig. 5. Training Performance of the *DropBot* system

These results suggest that the model achieved near-perfect accuracy in identifying battery objects within single-class scenarios, with extremely high confidence in detections. The slightly lower mAP@0.5:0.95 reflects minor localization imprecision at tighter Intersection-over-Union (IoU) thresholds, which is common in small-object detection tasks [16, 11].

YOLO-based vision systems have been widely applied in waste sorting and object detection, often achieving mAP values between 73–99% depending on task complexity and image conditions [4, 6]. In this context, *ZapSortBot*'s performance falls within the upper range of reported benchmarks, validating the suitability of YOLOv8 for battery classification in controlled lighting and orientation environments.

Furthermore, the system's robustness positions it as a viable alternative to high-cost spectroscopic or X-ray solutions for low- to mid-volume operations, as also advocated by Jose et al. [5] and Gadgil et al. [3], who emphasize scalable AI-driven approaches for battery sorting tasks. This visual distribution analysis supports the effectiveness of the training dataset and provides indirect validation of the model's ability to distinguish between classes based on spatial and dimensional cues.

2) Real life Classification

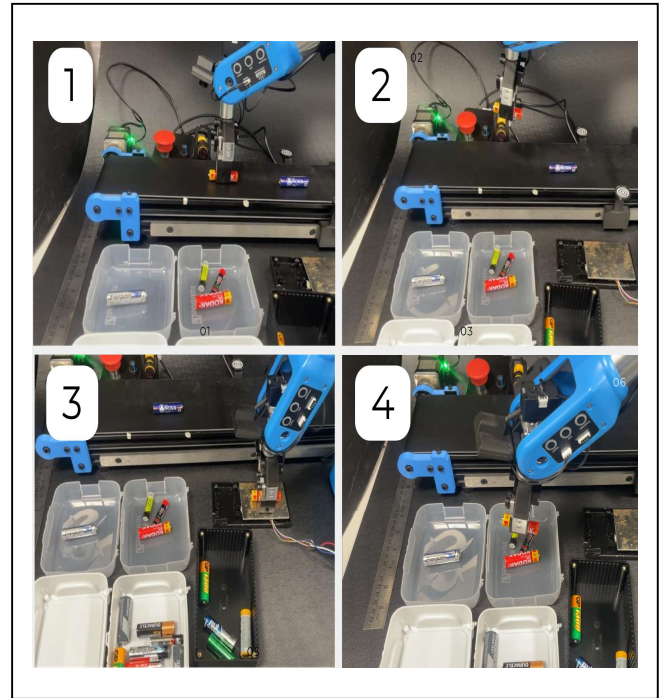


Fig. 6. Classification in progress

Figure 6 displays the system's classification in progress step by step. The 4 steps show the robot picking a detected battery up (1), displacing it (2), weighing it on the platform (3) and finally classifying it (4). The *ZapSortBot* system achieved 98.3% classification accuracy across all tested AA and AAA battery types, encompassing alkaline, zinc-carbon, NiMH, and lithium chemistries. Each size-chemistry combination was evaluated using 30 real life representative samples (150 batteries in total). No misclassifications were observed among batteries that adhered to expected physical characteristics (namely, standard weight ranges and identifiable color profiles).

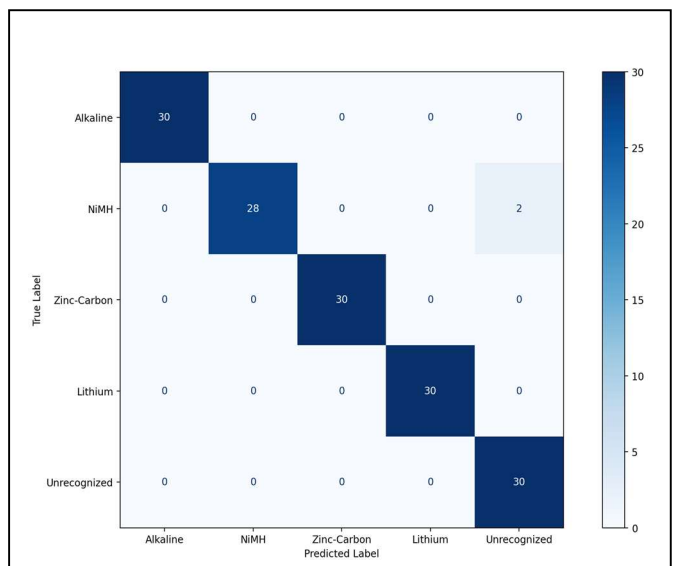


Fig. 7. Confusion Matrix

As represented in the confusion matrix in figure 7, two different green AA NiMH batteries with weights like alkaline cells were appropriately flagged as “unrecognized” due to their colour-weight-size combinations that don’t match any expected characteristics. This aligns with safety-first practices in automated sorting, where cautious classification reduces the risk of hazardous cross-contamination [8, 11]. Notably, once this edge case was added via the integrated UI and the model retrained, the battery was accurately classified (demonstrating *ZapSortBot*’s adaptability and real-world usability). This iterative refinement mirrors the hybrid reasoning strategies recommended for handling visually misleading or degraded batteries [10].

ZapSortBot demonstrated robust performance across a range of test conditions, including variable lighting and repeated operation cycles. Its classification accuracy remained stable, validating its readiness for real-time automation environments where input variability is common [2]. This robustness is driven by its hybrid inference pipeline, which combines HSV-based colour segmentation with weight thresholds (an approach shown in literature to outperform rigid, single-sensor systems in dynamic settings) [15, 3].

This result validates the system’s multi-stage classification pipeline, which combines real-time YOLOv8-based object detection, vision+weight-based chemistry/model inference. The system’s performance is consistent with other vision-based systems that report high accuracies when models are trained under well-controlled conditions with balanced datasets [16, 5].

While 98.3% accuracy is rare in real-world applications, this result highlights *ZapSortBot*’s reliability within its designed input domain. As noted in recent studies [9, 2], combining physical, visual, and AI-driven reasoning is essential for handling the diversity and ambiguity present in modern battery waste streams.

IV. CONCLUSION

This paper presented *ZapSortBot*, a low-cost, modular battery sorting system combining real-time computer vision, and weight inference to identify battery chemistry types. Designed for scalable deployment in settings where industrial solutions are impractical, *ZapSortBot* integrates YOLOv8 detection, and microcontroller-based weight checks to deliver high accuracy with minimal hardware and computational demands.

Compared to existing literature, which often emphasizes high-cost spectroscopy or X-ray-based systems [14, 13], *ZapSortBot* introduces a pragmatic alternative suitable for decentralized settings. Its hybrid logic approach (integrating physical and visual cues) mirrors techniques identified as effective in handling ambiguous or degraded batteries [10, 8], while the modular, update-ready design aligns with design-for-recyclability trends encouraged by recent regulations [12, 7].

Evaluation results confirm the system’s effectiveness: *ZapSortBot* achieved 98.3% classification accuracy on standard battery samples across AA and AAA form factors and major chemistries (alkaline, NiMH, zinc, lithium). Its

above 98% accuracy and real-time robustness demonstrate that it not only meets its functional goals but also offers a safe, reliable framework for scalable deployment in automated recycling workflows.

Looking forward, future work may explore the integration of low-cost spectroscopy or thermal sensors to expand the classification range. The introduction of self-updating edge-case recognition, improved dataset augmentation pipelines, and closed-loop robotic feedback could further enhance system generalization and autonomy.

All source code, datasets, and supplementary materials used in this work are publicly available at:

<https://github.com/ymbarkay/ZapSortbot.git>

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